SSL

```
postgres@bdd-VirtualBox:~$ psql
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1))
Type "help" for help.

postgres=#
```

*** Com podem veure fem una connexió psql i no es fa mitjançant SSL
comanda : **psql**

```
root@bdd-VirtualBox:/var/lib/postgresql/14/main# openssl genrsa -out server.key 2048
root@bdd-VirtualBox:/var/lib/postgresql/14/main# openssl req -new -x509 -days 365 -key server.key -out server.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:EU
State or Province Name (full name) [Some-State]:Barcelona
Locality Name (eg, city) []:Blanes
Organization Name (eg, company) [Internet Widgits Pty Ltd]:
Organizational Unit Name (eg, section) []:
Common Name (e.g. server FQDN or YOUR name) []:
Email Address []:
root@bdd-VirtualBox:/var/lib/postgresql/14/main# chmod 600 server.key
root@bdd-VirtualBox:/var/lib/postgresql/14/main#
```

*** Començem amb la generació del certificat i de la clau i li donem permisos
comanda1 : **genrsa -out server.key 2048**
comanda2 : **openssl req -new -x509 -days 365 -key server.key -out server.crt**
comanda3 : **chmod 600 server.key**
req : **sudo apt install openssl**

```
# - SSL -

ssl = on
#ssl_ca_file = ''
ssl_cert_file = '/var/lib/postgresql/14/main/server.crt'
#ssl_crl_file = ''
#ssl_crl_dir = ''
ssl_key_file = '/var/lib/postgresql/14/main/server.key'
#ssl_ciphers = 'HIGH:MEDIUM:+3DES:!aNULL' # allowed SSL ciphers
#ssl_prefer_server_ciphers = on
#ssl_ecdh_curve = 'prime256v1'
#ssl_min_protocol_version = 'TLSv1.2'
#ssl_max_protocol_version = ''
#ssl_dh_params_file = ''
#ssl_passphrase_command = ''
#ssl_passphrase_command_supports_reload = off
```

*** Entrem a l'arxiu de configuració de postgresql i anem a la secció de SSL, el posem en on, i el certificat i la clau que acabem de crear la posem

comanda : **sudo nano /etc/postgresql/14/main/postgresql.conf**

```
postgres@bdd-VirtualBox:/home/bdd$ psql -h 127.0.0.1 -U postgres
could not change directory to "/home/bdd": Permission denied
Password for user postgres:
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

postgres=#
```

*** Fem una connexió amb psql i com podem veure estem fent servir SSL
comanda : **psql -h 127.0.0.1 -U postgres**

```
  GNU nano 6.2                                                    /usr/local/bin/renovar-ssl.sh *
#!/bin/bash

# Configuración de rutas
DATA_DIR="/var/lib/postgresql/14/main"
LOG_FILE="/var/log/postgresql/renovacion_ssl.log"

echo "--- Iniciando renovación SSL: $(date) ---" >> $LOG_FILE

# 1. Generar nueva llave y certificado (válido por 1 año)
# Usamos -nodes para que no pida contraseña y el proceso sea automático
openssl req -new -x509 -days 365 -nodes -newkey rsa:2048 \
  -keyout $DATA_DIR/server.key \
  -out $DATA_DIR/server.crt \ >> $LOG_FILE 2>&1

# 2. Corregir permisos
chown postgres:postgres $DATA_DIR/server.key $DATA_DIR/server.crt
chmod 600 $DATA_DIR/server.key
chmod 644 $DATA_DIR/server.crt

# 3. Reiniciar PostgreSQL para aplicar los nuevos certificados
systemctl restart postgresql >> $LOG_FILE 2>&1

echo "--- Renovación completada con éxito ---" >> $LOG_FILE
```

*** Creem un script d'autorenovació de certificat de SSL, el creem a /usr/local/bin
comanda : **sudo nano /usr/local/bin/renovar-ssl. sh**

script :

renovar-ssl.sh

comanda2 : **sudo chmod +x /var/lib/postgresql/renovar_ssl.sh**

```
  GNU nano 6.2                                                    /tmp/crontab.p3gwQO/crontab *
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h  dom mon dow   command

0 3 1 * * /usr/local/bin/renovar-ssl.sh
```

*** Configurem cron per a que s’executi cada 6 mesos a les 1 de la nit aquest script
comanda1 : **sudo crontab -e**
comanda2 : **0 3 1 * * /var/lib/postgresql/renovar_ssl.sh > /var/log/postgresql/asistente_ssl.log 2>&1**

```
root@bdd-VirtualBox:/home/bdd# sudo /usr/local/bin/renovar-ssl.sh
root@bdd-VirtualBox:/home/bdd# cat /var/log/postgresql/renovacion_ssl.log
--- Iniciando renovación SSL: jue 16 abr 2026 18:53:34 CEST ---
req: Use -help for summary.
--- Renovación completada con éxito ---
root@bdd-VirtualBox:/home/bdd# su postgres
postgres@bdd-VirtualBox:/home/bdd$ psql -h 127.0.0.1 -U postgres
could not change directory to "/home/bdd": Permission denied
Password for user postgres:
psql (14.19 (Ubuntu 14.19-0ubuntu0.22.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

postgres=#
```

*** Executem l’script i mirem els logs per veure que funciona correctament, com podem veure funciona, fem la connexió i es mitjançant SSL
comanda : **sudo /usr/local/bin/renovar-ssl. sh**
comanda1 : **cat /var/log/postgresql/renovacion.ssl.log**
comanda2 : **psql -h 127.0.0.1 -U postgres**